

Metrics Match Consequences

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can apply a declared score threshold including equality.

I can compare false alarms and misses at two thresholds.

I can separate validation choices from final testing.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Metric choice depends on intended use, the positive class, who is affected, and the consequences of each error.



LEARNING TARGET A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What evidence is needed before choosing a threshold?

Builder name and experiment date:

Readiness for the connected explanation

Which denominator defines recall?

This week's end goal

By the end, I can explain metrics match consequences using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use metrics match consequences to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A classifier score is not automatically a probability. Declare blocked positive and predict blocked when score is at least the threshold. Changing this boundary changes decisions without necessarily retraining the model. Select it using development evidence and a task-relevant rule, not final-test answers.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines thresholds and consequences.

3. See how it works

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

Select metrics for screening



ML

MODEL

Metric choice depends on intended use, the positive class, who is affected, and the consequences of each error.



OUT

EVIDENCE

and forecasting tasks.

QUESTION Select metrics for screening, recommendation, and forecasting tasks.

MODEL IDEA Metric choice depends on intended use, the positive class, who is affected, and the consequences of each error.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Screening task treats missed positives as costly, so emphasize recall and human follow-up. Recommendation task may track precision among shown items plus user benefit.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Six records have score/reference pairs A .9/blocked, B .7/clear, C .6/blocked, D .4/blocked, E .3/clear, F .1/clear. At threshold .5, TP=2, FP=1, FN=1, TN=2. At .3, TP=3, FP=2, FN=0, TN=1. Equality is positive, so E is flagged at .3.

```
scores = [0.9, 0.7, 0.6, 0.4, 0.3, 0.1]
actual = [True, False, True, True, False, False]

for threshold in [0.5, 0.3]:
    predicted = [s >= threshold for s in scores]

    fp = sum(p and not a for p, a in zip(predicted, actual))
    fn = sum(not p and a for p, a in zip(predicted, actual))
    print(threshold, fp, fn, fp + 4 * fn)

# Output rows: 0.5 1 1 5, then 0.3 2 0 2.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Connect the picture and the explanation

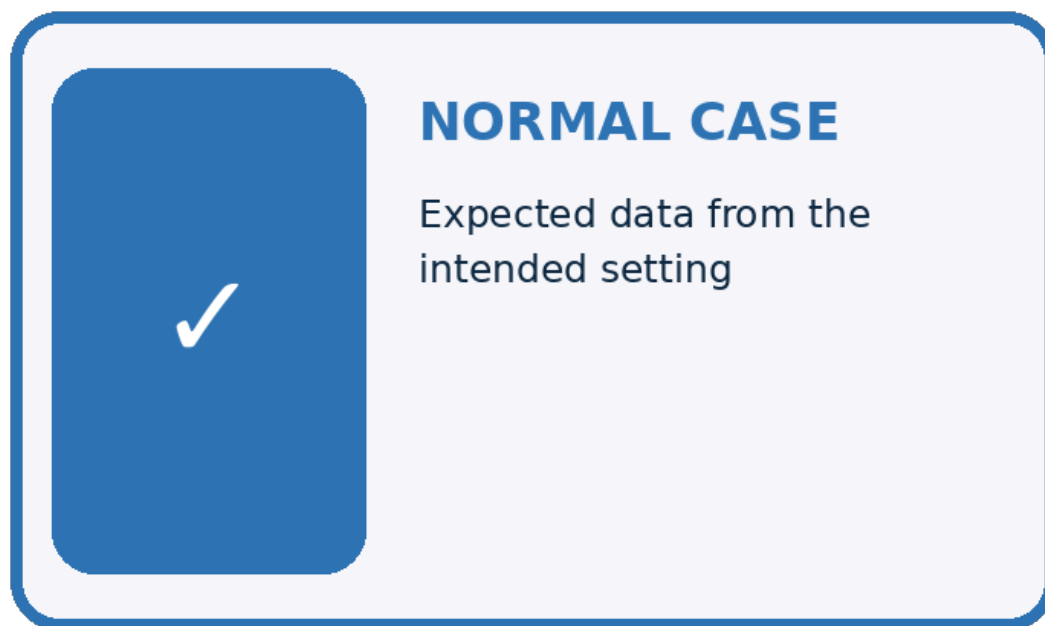
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





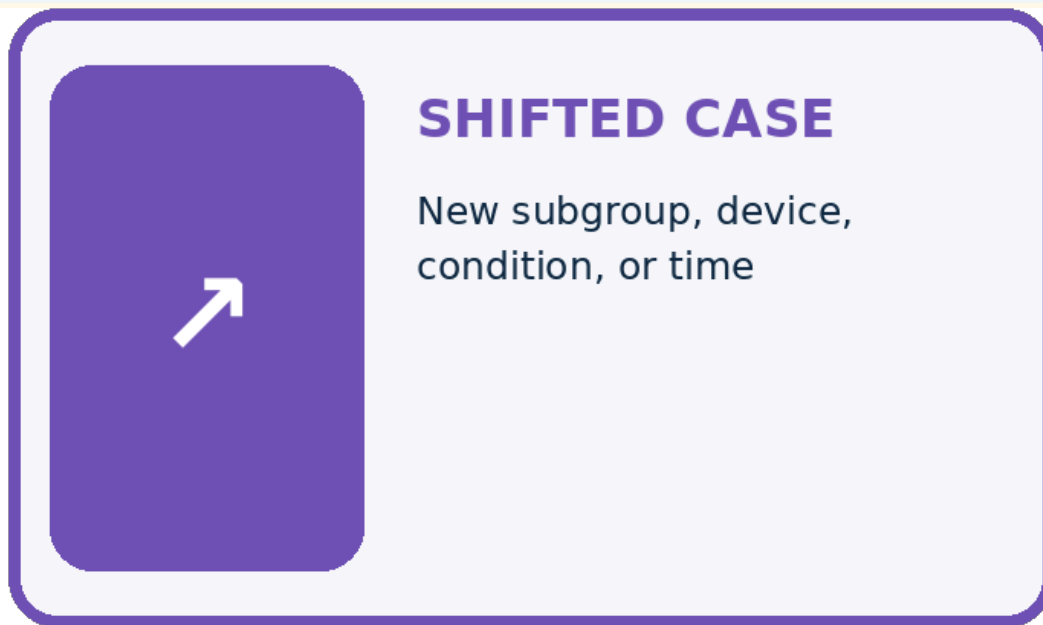
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Miss and false-alarm costs swap to 1 and 5.

Prediction: Threshold 0.7 costs 1 while 0.4 costs 5; preferred threshold switches.

Reason: Metric choice encodes the declared consequences, not a universal best threshold.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

CONSEQUENCE

real effect of a prediction error

2

THRESHOLD

score boundary for a class decision

3

COST MATRIX

relative impact assigned to error types

4

METRIC SUITE

multiple measures covering distinct concerns

Term · Precise meaning · My ML example

Term: consequence

Precise meaning: real effect of a prediction error

My ML example: _____

Term: threshold

Precise meaning: score boundary for a class decision

My ML example: _____

Term: cost matrix

Precise meaning: relative impact assigned to error types

My ML example: _____

Term: metric suite

Precise meaning: multiple measures covering distinct concerns

My ML example: _____

MISCONCEPTION TO REPAIR The same highest-accuracy metric should choose every model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Score

Meaning: A model output used to rank or decide

Word: Threshold

Meaning: A boundary that converts a score to a label

Word: False alarm

Meaning: A positive prediction for a negative reference

Word: Cost rule

Meaning: A declared way to value different mistakes

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 25 • WORKED EVIDENCE

- 1 Screening task treats missed positives as costly, so emphasize recall and human follow-up.
- 2 Recommendation task may track precision among shown items plus user benefit.
- 3 Height forecast uses MAE in centimeters and residual slices.
- 4 No metric replaces a written consequence analysis and subgroup review.

STEP 1 Screening task treats missed positives as costly, so emphasize recall and human follow-up.

STEP 2 Recommendation task may track precision among shown items plus user benefit.

STEP 3 Height forecast uses MAE in centimeters and residual slices.

STEP 4 No metric replaces a written consequence analysis and subgroup review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Thresholds and consequences at a glance

Part · Evidence or action · Interpretation

Part: Threshold .5

Evidence or action: One false alarm one miss

Interpretation: Four of six correct

Part: Threshold .3

Evidence or action: Two false alarms no misses

Interpretation: Four of six correct

Part: Decision rule

Evidence or action: Include equality

Interpretation: Record before evaluation

Lowering the threshold here finds D but adds E as a false alarm. The two accuracies are equal while consequences differ. Precision is 2/3 at .5 and 3/5 at .3; recall rises from 2/3 to 1. This behavior follows these fixed scores and records, not a promise of future performance.

Give TP, FP, FN, TN at both thresholds. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Metrics Match Consequences

Translate a threshold choice into missed-leaf and false-alarm costs.

The facts we will use

Scores/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at $\text{score} \geq \text{threshold}$. Miss cost 5; false-alarm cost 1.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Metrics Match Consequences

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Threshold0.7

TP 1, FN 1, FP 0, TN 2: cost5.



2. Threshold0.4

TP 2, FN 0, FP 1, TN 1: cost1.



3. Choose for stated costs

Threshold0.4 has smaller validation cost on these four cases.

CHANGE ONE THING

Miss and false-alarm costs swap to 1 and 5.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Threshold0.7

What happens and why: TP 1, FN 1, FP 0, TN 2: cost5.

Step: 2. Threshold0.4

What happens and why: TP 2, FN 0, FP 1, TN 1: cost1.

Step: 3. Choose for stated costs

What happens and why: Threshold0.4 has smaller validation cost on these four cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Threshold0.4 has smaller validation cost on these four cases. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Toy costs illustrate decision design; real use needs agreed consequences, review capacity and fresh evaluation.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → **Diagnose** → **Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: The same highest-accuracy metric should choose every model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · **Expected** · **Actual** · **Likely cause / next test**

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE The same highest-accuracy metric should choose every model.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

These invented costs support arithmetic practice, not real robot deployment. Stakeholders must justify actual consequences. Record every tried threshold and retain a separate final test; a threshold is part of the selected method.

Why not select the threshold with the lowest final-test cost?

8. Practise together

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Suppose the classroom cost is one point per false alarm and four per miss. Total costs are 5 at .5 and 2 at .3, so .3 wins among these two candidates under this rule. If false alarms cost four and misses cost one, costs are 5 and 8, reversing the choice.

Practise with less help: Count the consequences

Use workbook W2 for the connected example “Thresholds and consequences”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Thresholds and consequences. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which labels determine the false-alarm count at threshold 0.4? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Metric Council: map stakeholders and error costs, select a metric suite, and defend what each metric cannot show.

1. Write the hypothesis and expected evidence before running.
2. Metric Council: map stakeholders and error costs, select a metric suite, and defend what each metric cannot show.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

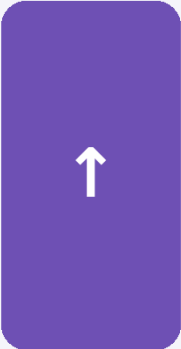
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: consequence, threshold, cost matrix, metric suite.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The same highest-accuracy metric should choose every model.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Approve the Plant Classifier metric suite: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

THIS WEEK'S CHECKPOINT Approve the Plant Classifier metric suite: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Select metrics for screening, recommendation, and forecasting tasks.

REQUIRED PROOF Select metrics for screening, recommendation, and forecasting tasks.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Reproducible Experiments.

13. Study a complete case

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

The question

Connect metrics to consequences.

Worked explanation

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using consequences, review capacity and validation evidence.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Connect error types to consequences

Evidence or stage · Value or result

Evidence or stage: False negative

Value or result: Spotted leaf missed

Evidence or stage: False positive

Value or result: Healthy leaf creates an alert

Evidence or stage: Recall

Value or result: Fraction of positives found

Evidence or stage: Precision

Value or result: Fraction of alerts correct

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

Connect metrics to consequences.

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using consequences, review capacity and validation evidence.

14. Try a new case independently

Independent case: Apply independently

Scores/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare thresholds .5 and .2 using the same cost.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Missing a diseased plant is considered more costly than a false alarm, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Approve the Plant Classifier metric suite: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What evidence is needed before choosing a threshold?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.